

SEC P vs NP log 2026-04-29

SEC (autonomous) — attributed to Ludovico Kubler

2026-04-29 00:40 UTC

SEC P vs NP — daily log 2026-04-29

3 cycles recorded

GF(2)-Rank Defect of Term-Indicator Matrix Lower-Bounds Monotone DNF for k-CLIQUE

- **Verdict:** INCONCLUSIVE
- **Bridge:** Matroid theory — the GF(2) vector matroid M_F realized by the term-indicator matrix $A_F \in \mathbb{F}_2^{m \times N}$ of a monotone DNF $F = \bigvee_{i=1}^m T_i$ (rows $\{T_i\}$, columns indexed by variables). Its rank $\rho_2(F) := \text{rank}_{\mathbb{F}_2}(A_F)$ is the canonical computable matroid invariant in the Whitney-Welsh tradition; the corresponding ‘rank defect’ $\Delta(F) := \log_2(m(F)+1) - \log_2(\rho_2(F)+1)$ is a polymatroid-style quantity (subadditive log-rank vs. log-cardinality) that is rarely deployed in monotone circuit lower bounds (Razborov 1985 used Bollobás set-pair / sunflower closure, not GF(2) span of clutter indicators). $\times$ Monotone DNF representation size of the k-CLIQUE_n indicator on $N = C(n,2)$ edge variables; equivalently, the minimum-cardinality clutter of minimal terms whose union gives k-CLIQUE_n.
- **Recorded:** 2026-04-29 00:11 UTC
- **Entry ID:** 023dd98c00ee

Statement

Let F be a monotone DNF on N variables with minimal-term family $H_F = \{T_1, \dots, T_m\}$, A_F its $m \times N$ indicator matrix, and define $\Delta(F) := \log_2(m+1) - \log_2(\text{rank}_{\mathbb{F}_2}(A_F)+1)$. Conjecture: (a) $\Delta(F \wedge G) \leq \Delta(F) + \Delta(G) + 1$ and $\Delta(F \vee G) \leq \max(\Delta(F), \Delta(G)) + 1$ (approximate submodularity through Boolean operations); (b) $\Delta(F) \leq \log_2(N+1)$ for any monotone DNF with $m \leq \text{poly}(N)$ (since $\text{rank}_{\mathbb{F}_2}(A_F) \geq 1$ and $m \leq \text{poly}(N)$); (c) for the canonical k-CLIQUE_n DNF $F^{\wedge\{n,k\}}$ with $k = \lfloor n/2 \rfloor$, $\Delta(F^{\wedge\{n,k\}}) \geq n/2 - 3 \cdot \log_2 n - 4$ for all $n \geq 6$, witnessing a $\Omega(n)$ gap between log-cardinality and log-GF(2)-rank.

Rationale

The GF(2) span of clique-edge-indicator vectors collapses to a sub-quadratic-rank object ($\text{rank} \leq C(n,2)$), while the minimal-term count $C(n, \lfloor n/2 \rfloor) \approx 2^n / \sqrt{n}$ is exponential — so the LOG ratio is $\Theta(n)$, much larger than any rank deficit attainable by poly-size clutters. Razborov’s approximation method tracks subadditive measures across AND/OR gates; an honest submodular Δ that is forced to $\Omega(n)$ on the clique clutter is exactly the missing ingredient for a Razborov-style argument that does not require sunflower closures, since GF(2)-rank is robust under union operations on supports. The bridge from matroid representability to monotone DNF size is concrete (one Gaussian elimination per gate), avoiding the natural-proofs / algebrization barriers that block spectral or Fourier approaches.

Novelty

- Judge: NOVEL over 11 arXiv hits

Top hits consulted: - [1806.02734v3] Spectral lower bounds for the orthogonal and projective ranks of a graph - [1801.08709v2] Adaptive Lower Bound for Testing Monotonicity on the Line - [1007.1875v2] Lower Bounds for Quantum Oblivious Transfer - [1904.05483v2] Parallels Between Phase Transitions and Circuit Complexity? - [2210.12983v2] Poisson multi-Bernoulli mixture filter with general target-generated measurements and arbitrary clutter

Empirical Test

- exit code: 1, elapsed: 0.10s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8b7aab2a.py", line 118, in <module>
    result = run_trial(seed)
            ^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8b7aab2a.py", line 78, in run_trial
    F_random = random_poly_dnf(n, n, 2) + random_poly_dnf(n, n**2, 3) + random_poly_dnf(n, n * math.floor(n/2), 1)
            ^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8b7aab2a.py", line 56, in random_poly_dnf
    selected_edges = random.sample(edge_set, w)
                    ^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/usr/lib/python3.12/random.py", line 413, in sample
    raise TypeError("Population must be a sequence. ")
```

TypeError: Population must be a sequence. For dicts or sets, use sorted(d).

Judge reasoning

The test crashed with a `TypeError` in `random_poly_dnf` (`random.sample` called on a non-sequence) before any data was produced, so neither the support nor falsification conditions could be evaluated. | next: Fix `random_poly_dnf` by passing a list (e.g., `list(edge_set)` or `sorted(edge_set)`) to `random.sample` and rerun the 30-seed sweep.

Coarse-Equivalence Invariance of Protocol-Pullback Multiplicity Across Bit-Permuted Gadgets

- **Verdict:** INCONCLUSIVE
- **Bridge:** Coarse Geometric Lifting (CGL) — Coarse Geometry / Roe Algebras (within Geometric Group Theory) $\times$ Deterministic 2-party communication complexity, specifically the multiplicity m_Π of the RoeCover extracted from a deterministic protocol's ProtocolPullback for the lifted function $f \circ G^n$
- **Recorded:** 2026-04-29 00:24 UTC
- **Entry ID:** aa9fe4edbf5

Statement

Let $G_1 = (X, Y, g, d)$ be a MetricGadget on small alphabet ($|X|, |Y| \leq 4$) and let G_2 be obtained from G_1 by a bi-Lipschitz bijection $\varphi: X \times Y \rightarrow X \times Y$ with distortion ≤ 2 (a coarse equivalence: φ permutes inputs while distorting d by at most a factor 2). Fix any boolean $f: \{0,1\}^n \rightarrow \{0,1\}$ with $n \leq 4$. Then for every deterministic protocol Π_1 computing $f \circ G_1^n$, there exists a deterministic protocol Π_2 computing $f \circ G_2^n$ with cost $\text{cost}(\Pi_2) \leq \text{cost}(\Pi_1) + O(n)$, and the multiplicities of their CoarsePullback covers at scale $R = \text{diam}(G)$ satisfy $m_{\{\Pi_2\}} \leq 2 \cdot m_{\{\Pi_1\}} + 1$. Equivalently, $\text{CLC}(f, G_1)$ and $\text{CLC}(f, G_2)$ agree up to a constant factor independent of f , instantiating Axiom A1

(Coarse-Lift Functoriality) at the level of the multiplicity invariant that A2 connects to communication complexity.

Rationale

This conjecture directly tests Axiom A1 (Coarse-Lift Functoriality) by asserting that the *multiplicity invariant* m_Π exposed by Axiom A2 (Protocol $\rightarrow$ Cover) is preserved under coarse equivalences of the underlying gadget. If A1 holds, a bi-Lipschitz ϕ on $X \times Y$ lifts coordinatewise to a bi-Lipschitz map on $(X \times Y)^n$ with the $d_\oplus$ metric (distortion preserved since $\square^1$ products of bi-Lipschitz maps remain bi-Lipschitz), so any Roe-style cover at scale R for the G_1 -lift can be transported to a cover at scale $\leq 2R$ for the G_2 -lift while at most doubling the multiplicity (each transported set may merge with one neighbor at the boundary, contributing the $+1$ slack). The $O(n)$ cost overhead in Π_2 reflects simulating ϕ coordinatewise. Verifying this empirically would falsify A1 if multiplicities diverge, and would strengthen confidence that CLC is genuinely a coarse invariant — a prerequisite for using it as a lower-bound technique against communication complexity.

Novelty

- Judge: NOVEL over 9 arXiv hits

Top hits consulted: - [2408.07701v2] A categorical interpretation of Morita equivalence for dynamical von Neumann algebras - [2312.08907v2] Roe algebras of coarse spaces via coarse geometric modules - [1706.00276v1] Counting coarse subsets of a countable group - [2110.02008v2] Lifted Reed-Solomon Codes and Lifted Multiplicity Codes - [1903.06084v8] Lifting coarse homotopies

Empirical Test

- exit code: 1, elapsed: 0.11s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8a94fc38.py", line 113, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8a94fc38.py", line 49, in run_trial
    phi_perms = generate_bi_lipschitz_permutations()
                 ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8a94fc38.py", line 35, in generate_bi_l
    if is_bi_lipschitz(phi):
       ^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8a94fc38.py", line 24, in is_bi_lipschi
    if dist(phi[(x1, y1)], phi[(x2, y2)]) > distortion * dist((x1, y1), (x2, y2)):
       ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8a94fc38.py", line 18, in dist
    return max(abs(ai - bi) for ai, bi in zip(a, b))
               ^^^^^^^^^
```

TypeError: 'int' object is not iterable

Judge reasoning

The test crashed with a `TypeError` in the bi-Lipschitz check (`dist()` called on integers instead of tuples) before any trials produced data, so neither the support nor falsification criteria can be evaluated. | next: Fix the `dist()` helper to handle the (x,y) pair representation correctly (ensure ϕ maps tuples to tuples and `dist` unpacks both arguments), then rerun the 5-seed sweep.

Persistent H_0 Component Count of Sign-Quotient Row Set Caps Real Rank

- **Verdict:** INCONCLUSIVE
- **Bridge:** Persistent homology / Vietoris-Rips topology of finite metric spaces (Vietoris 1927; Edelsbrunner-Letscher-Zomorodian 2002; Carlsson-Zomorodian 2005; Carlsson 2009). The 0-th persistent Betti number $\beta_0(\text{VR}(X, \varepsilon))$ of a finite metric space (X, d) at scale ε is the number of connected components of the ε -thresholded distance graph, computable by Kruskal/union-find in $O(|X|^2 \alpha(|X|))$. Rarely deployed in communication complexity (TDA literature is dominated by data-analysis applications; we are aware of <5 papers connecting VR persistence of communication matrices to log-rank / D^{cc}). $\times$ Real rank $\text{rk}_R(M_f)$ of the $2^k \times 2^k \pm 1$ communication matrix $M_f[x, y] = (-1)^{f(x, y)}$ of a Boolean function $f: \{0, 1\}^k \times \{0, 1\}^k \rightarrow \{0, 1\}$ — the canonical Lovász log-rank / Mehlhorn-Schmidt target for deterministic two-party communication complexity $D^{\text{cc}}(f) \geq \log_2 \text{rk}_R(M_f)$.
- **Recorded:** 2026-04-29 00:40 UTC
- **Entry ID:** bc750201aaf2

Statement

Let $X_f^{\pm} \subset \{\pm 1\}^{2^k} / \{\pm 1\}$ be the set of distinct rows of M_f modulo global sign, equipped with the projective Hamming distance $d_{\pm}(r, s) = \min(d_H(r, s), 2^k - d_H(r, s))$. Let $\tau_{\pm}(M_f) := \beta_0(\text{VR}(X_f^{\pm}, 2^{\{k-2\}}))$, the number of components of the graph linking r, s whenever $d_{\pm}(r, s) \leq 2^{\{k-2\}}$. Then for every Boolean f and every $k \in \{3, 4, 5, 6\}$, $\tau_{\pm}(M_f) \leq \text{rk}_R(M_f)$; a single (k, f) with $\tau_{\pm}(M_f) > \text{rk}_R(M_f)$ refutes.

Rationale

Rank- $r \pm 1$ matrices have all rows lying in an r -dimensional subspace of $\mathbb{R}^N$, and ± 1 vectors in such a subspace are constrained to a structured (centrally-symmetric) finite set whose pairwise distances avoid the ‘mid-Hamming’ regime around $N/2$ only by sitting in a small affine pattern; persistence at scale $N/4$ should therefore not be able to manufacture more than r distinct components after the $\pm$ -quotient. This bridges the topological ‘shape-counting’ of the row metric space to the algebraic rank-bound side of log-rank, a structural inequality complementary to spectral approaches and untouched by natural-proofs (rk_R is the property being lower-bounded, not a hardness predicate on truth tables) and untouched by algebrization (no oracle relativization is used).

Novelty

- Judge: NOVEL over 5 arXiv hits

Top hits consulted: - [LIPIcs.ESA.2024.29] A Euclidean Embedding for Computing Persistent Homology with Gaussian Kernels - [s2:609241bc94e187c943bd6d4ec6bcf0c1ababa] Topology in sensor networks - [s2:2505.16879] How high is ‘high’? Rethinking the roles of dimensionality in topological data analysis and manifold learning - [PL00009265] Harmonic Analysis, Real Approximation, and the Communication Complexity of Boolean Functions - [s2:f23504fcde864a63a36bc0be33c3a3f97d615] The Nature of Migration and Its Impact on Families in Peru

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7aad0d36.py", line 185, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7aad0d36.py", line 163, in run_trial
    M.append([Fraction(row[i]) for i in range(n)])
```

~~~~^^^

IndexError: tuple index out of range

### **Judge reasoning**

The test crashed with an IndexError before producing any RESULT line, so neither the support nor the falsification condition could be evaluated. Per rule 2, a crashed test yields INCONCLUSIVE. | next: Fix the row-construction bug (tuple index out of range when building M from row entries) and re-run the full  $4 \times 30 \times 6$  grid to obtain  $\tau_{\pm}$  and rk\_R counts.